PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

AI for Early Detection of Heart Disease using ECG and Imaging

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Nishtha (2210991993)

Prachi (2210992052)

Raghav (2210992116)

Sehdev Pratap(2210992278)

Supervised By:

Dr. Gifty Gupta

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

S.No.	Title	Page No.
	Declaration	3
	Acknowledgement	4
	Abstract	5
1	Introduction	6-7
2	Methodology	7-8
3	Tools and Technologies	9-10
4	Implementation	10-11
5	Major Findings/Outcomes/Output/Results	12-13
6	Conclusion and Future Scope	13-14
7	References	14-15

DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"AI for Early Detection of Heart Disease using ECG and Imaging"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Dr. Gifty Gupta. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 29th April, 2026

Name of the Students :

Nishtha (2210991993)

Prachi (2210992052)

Raghav (2210992116)

Sehdev Pratap (2210992278)

This is to certify that the above statement made by the candidates are correct to the best of our knowledge and belief.

Name of the Mentor: Dr. Gifty Gupta
Designation : Assistant Professor
Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India

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Name: Nishtha
Roll No: 2210991993

Name: Prachi
Roll No: 2210992052

Name: Raghav
Roll No: 2210992116

Name: Sehdev Pratap
Roll No: 2210992278

Abstract :

Cardiovascular diseases remain one of the leading causes of mortality worldwide, making early detection and accurate diagnosis a critical challenge in modern healthcare systems. Electrocardiogram (ECG) analysis is a widely used, non-invasive, and cost-effective technique for monitoring heart activity and identifying potential abnormalities. However, traditional ECG interpretation largely depends on expert cardiologists, which can be time-consuming, subjective, and prone to human error, especially in high-demand clinical environments.

This project presents a deep learning-based approach for the early detection and classification of heart diseases using ECG image data. The proposed system leverages a **ResNet50-based convolutional neural network architecture** combined with transfer learning to effectively extract and learn complex features from ECG images. The model is designed to perform multi-class classification across five categories: AHB (Atrioventricular Heart Block), COVID-19-related cardiac conditions, HMI (History of Myocardial Infarction), MI (Myocardial Infarction), and Normal cases.

A dataset consisting of 4568 ECG images is utilized, which is divided into training, validation, and testing subsets. Several preprocessing techniques, including image resizing, normalization, and data augmentation, are applied to improve model performance and generalization. The pretrained ResNet50 model is used as a feature extractor, while additional fully connected layers are incorporated to adapt the model specifically to the ECG classification task.

The proposed model demonstrates strong performance, achieving an overall accuracy of **87.70%**, along with robust evaluation metrics such as precision, recall, F1-score, and ROC-AUC values ranging from **0.87 to 0.97** across different classes. The training and validation curves indicate stable convergence with minimal overfitting, highlighting the model's reliability and generalization capability.

Overall, the results suggest that the integration of deep learning techniques with ECG image analysis can significantly enhance the accuracy and efficiency of heart disease detection. The proposed system has the potential to serve as a supportive tool for clinical decision-making, reducing diagnostic time and assisting healthcare professionals in early intervention and treatment planning.

1. Introduction :

1.1 Background of the Study :

Cardiovascular diseases (CVDs) are one of the leading causes of mortality worldwide, accounting for a significant number of deaths every year. Early detection and timely diagnosis play a crucial role in reducing fatality rates and improving patient outcomes. Among various diagnostic techniques, the Electrocardiogram (ECG) is widely used as a non-invasive and cost-effective method for monitoring the electrical activity of the heart.

Traditionally, ECG analysis is performed manually by cardiologists, which can be time-consuming, subjective, and prone to human error. With the increasing number of patients and limited availability of medical experts, accurate and timely diagnosis becomes a challenging task. Additionally, subtle abnormalities in ECG patterns can sometimes be difficult to detect through manual observation.

With advancements in artificial intelligence, particularly deep learning, automated systems have emerged as powerful tools for medical image analysis. Convolutional Neural Networks (CNNs) can extract complex features from ECG images and assist in accurate classification of cardiac conditions. Therefore, there is a growing need for intelligent systems that can support early detection of heart diseases using ECG data, improving diagnostic efficiency and reliability.

1.2 Problem Statement :

Traditional methods for diagnosing heart diseases rely heavily on manual interpretation of ECG reports by medical professionals. This process is not only time-intensive but also subject to variability depending on the expertise of the clinician. In high-demand healthcare environments, this can lead to delays in diagnosis and increased chances of misinterpretation.

Existing machine learning approaches have shown some improvement in classification tasks; however, they often struggle with complex ECG patterns and limited datasets. Moreover, many systems lack the ability to effectively generalize across multiple cardiac conditions, especially in multi-class classification scenarios.

Therefore, there is a need for an efficient, automated, and accurate system that can analyze ECG images and classify different types of heart conditions. Such a system should be capable of handling imbalanced datasets, extracting meaningful features, and providing reliable predictions to assist healthcare professionals in early diagnosis and treatment planning.

1.3 Objectives of the Study :

The primary objectives of this project are:

- To develop a deep learning-based system for heart disease detection using ECG images
- To perform multi-class classification of cardiac conditions
- To utilize transfer learning with ResNet50 for improved feature extraction
- To evaluate model performance using metrics such as accuracy, precision, recall, and ROC-AUC
- To minimize misclassification and improve diagnostic reliability
- To support early detection and assist in clinical decision-making

1.4 Scope of the Project :

This project focuses on:

- ECG image-based heart disease classification
- Application of deep learning techniques (ResNet50)
- Multi-class classification of cardiac conditions
- Performance evaluation using standard metrics
- Analysis of training and validation results

It excludes:

- Real-time patient monitoring systems
- Integration with IoT medical devices
- Direct clinical deployment without validation
- Analysis of non-image medical data (e.g., clinical reports, lab results)

2. Methodology :

2.1 Dataset Description

The dataset used in this project is a publicly available ECG image dataset consisting of 4568 images. These images represent different cardiac conditions and are categorized into five classes:

- AHB (Atrioventricular Heart Block)
- COVID-19 related cardiac condition
- HMI (History of Myocardial Infarction)
- MI (Myocardial Infarction)
- Normal

The dataset is divided into:

- 70% Training Data

- 20% Testing Data
- 10% Validation Data

The dataset is imbalanced, which makes classification more challenging and requires careful handling during model training.

2.2 Data Preprocessing

To ensure consistency and improve model performance, several preprocessing steps were applied:

- All images were resized to 224 × 224 pixels to match the input requirements of the ResNet50 model.
- Pixel values were normalized to the range [0,1] to improve training stability.
- Data augmentation techniques were applied to enhance generalization:
    - Random rotation (±15 degrees)
    - Zoom transformation
    - Horizontal and vertical flipping

These steps help in increasing dataset diversity and reducing overfitting.

2.3 Model Architecture

The proposed model is based on the ResNet50 architecture, a deep convolutional neural network that utilizes residual learning to improve training efficiency and overcome the vanishing gradient problem.
A pretrained ResNet50 model, trained on the ImageNet dataset, is used as a feature extractor. The base layers are initially frozen to preserve learned features, and a custom classification head is added on top.

The classification head consists of:

- A Global Average Pooling layer to reduce feature dimensions
- A Dense layer with 256 neurons using ReLU activation
- A Dropout layer with a rate of 0.5 to prevent overfitting
- A final Dense layer with Softmax activation for multi-class classification

The model takes input images of shape (224, 224, 3) and outputs probability scores for each class.

2.4 Training Configuration

The model was trained using the Adam optimizer with a learning rate of 0.0001, which ensures efficient and adaptive learning. A batch size of 32 was used, and the model was trained for 50 epochs. Categorical Cross-Entropy was used as the loss function since the problem involves multi-class classification. GPU acceleration (NVIDIA T4/P100) was utilized to speed up the training process. The model performance was evaluated using multiple metrics, including accuracy, precision, recall, F1-score, and ROC-AUC, to ensure reliable and comprehensive assessment.
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3. Tools and Technologies :

3.1 Programming Language :

Python

- Deep learning model development
- Easy integration with ML frameworks
- Efficient data handling and preprocessing

3.2 Libraries and Frameworks :

TensorFlow / Keras

- Model building and training
- Deep learning architecture implementation
- GPU acceleration support

ResNet50 (Pretrained Model)

- Feature extraction from ECG images
- Transfer learning implementation
- Improved classification performance

NumPy

- Numerical computations
- Image data processing
- Array and matrix operations

- **Matplotlib**
- Visualization of results
- Plotting accuracy and loss graphs
- Performance analysis

- **Scikit-learn**
- Evaluation metrics calculation
- Confusion matrix generation
- ROC curve analysis

3.3 Machine Learning Techniques :

- Transfer learning using ResNet50
- Convolutional Neural Networks (CNN)
- Multi-class classification
- Data augmentation for generalization
- Dropout for overfitting prevention

3.4 Hardware Requirements :

- Computer/Laptop
- GPU support (NVIDIA T4/P100)
- Minimum 8GB RAM
- Storage for dataset and model

4. Implementation :

4.1 Data Acquisition

The ECG image dataset used in this project is obtained from publicly available sources and consists of labeled images representing different cardiac conditions. The dataset is organized into separate directories based on class labels, ensuring efficient data management and accessibility during model training.

Each image corresponds to one of the five categories: AHB, COVID-19, HMI, MI, or Normal. The dataset is divided into training, validation, and testing subsets to enable proper learning, tuning, and unbiased evaluation of the model.

4.2 Data Preprocessing

Before feeding the data into the model, several preprocessing steps are performed to ensure consistency and improve performance.

All ECG images are resized to **224 × 224 pixels** to match the input requirements of the ResNet50 architecture. Pixel values are normalized to the range **[0,1]**, which helps stabilize the training process and improves convergence.

Data augmentation techniques such as rotation, zooming, and flipping are applied to increase dataset diversity and reduce overfitting. These transformations enable the model to learn robust and generalized features.

4.3 Model Development

The model is developed using the **ResNet50 architecture** with transfer learning. The pretrained ResNet50 model, trained on the ImageNet dataset, is used as a feature extractor.

The base layers of the model are frozen to retain previously learned features, while a custom classification head is added on top. This classification head includes a Global Average Pooling layer, a Dense layer with ReLU activation, a Dropout layer to prevent overfitting, and a final Softmax layer for multi-class classification.

4.4 Model Training

The model is trained using the Adam optimizer with a learning rate of 0.0001 and categorical cross-entropy as the loss function. Training is performed over multiple epochs with a batch size of 32.

During the training process, both training and validation performance are monitored. Accuracy and loss curves are analyzed to ensure proper learning and to detect overfitting. GPU acceleration is utilized to enhance computational efficiency and reduce training time.

4.5 Model Evaluation and Prediction

After training, the model is evaluated on the test dataset to assess its performance.

Accuracy is calculated to measure overall effectiveness. A confusion matrix is generated to analyze class-wise predictions. Additional metrics such as precision, recall, and F1-score are computed to provide a detailed evaluation of the model's performance. ROC curves are also plotted to assess the classification capability across different classes.

4.6 System Workflow

1. Dataset is loaded and organized into classes
2. Images are resized and normalized
3. Data augmentation techniques are applied
4. ResNet50 model is initialized with pretrained weights
5. Custom classification layers are added
6. Model is trained using training dataset
7. Validation is performed during training
8. Model is evaluated on test dataset
9. Performance metrics are calculated
10. Final predictions are generated

5. Results

The experimental evaluation of the proposed deep learning framework demonstrated strong performance in the multi-class classification of ECG images for early heart disease detection. The ResNet50-based model achieved an overall classification accuracy of **87.70%**, indicating its effectiveness in identifying different cardiac conditions from ECG image data. Compared to traditional machine learning approaches, the deep learning model showed improved feature extraction capability and better generalization across multiple classes, highlighting the robustness of transfer learning in medical image analysis.

One of the key outcomes observed was the model's stable training behavior and convergence. The training and validation accuracy curves exhibited a consistent upward trend, reaching approximately **94% training accuracy** and **89% validation accuracy** over 50 epochs. The close alignment between these curves indicates minimal overfitting and strong generalization capability. Additionally, the loss curves showed a steady decrease during training, confirming efficient optimization and reduced classification error throughout the learning process.

The model also demonstrated reliable performance across individual classes, as reflected in the confusion matrix and classification metrics. Precision, recall, and F1-scores were balanced for most classes, indicating consistent predictive performance. However, slight variations were observed in certain categories due to dataset imbalance and similarities in ECG patterns. Despite these challenges, the model maintained satisfactory classification accuracy, showing its ability to handle complex and imbalanced medical datasets.

Further analysis using Receiver Operating Characteristic (ROC) curves confirmed the strong discriminative capability of the model. The Area Under Curve (AUC) values ranged from **0.87 to 0.97** across the five classes, with the highest performance observed in the COVID-19 and MI categories. The ROC curves remained well above the diagonal baseline, indicating high sensitivity and specificity in distinguishing between different cardiac conditions.

Another important observation was the computational efficiency of the model. The use of transfer learning with ResNet50 significantly reduced training time while maintaining high performance. The model was able to process and classify ECG images efficiently using GPU acceleration, making it suitable for practical applications in healthcare environments. Memory utilization and processing requirements remained manageable, allowing deployment on standard systems with moderate computational resources.

Overall, the results confirm that the proposed framework provides a reliable, accurate, and efficient solution for automated ECG image classification. The integration of deep learning techniques enables effective early detection of heart diseases, reduces dependency on manual interpretation, and supports clinical decision-making. The system demonstrates strong potential for real-world applications in medical diagnostics, offering a scalable and cost-effective approach to improving cardiovascular healthcare outcomes.

6. Conclusion and Future Scope :

6.1 Conclusion

The proposed system successfully demonstrates the application of deep learning techniques for the early detection and classification of heart diseases using ECG image data. By leveraging the ResNet50 architecture with transfer learning, the model effectively extracts meaningful features and performs accurate multi-class classification across different cardiac conditions. The achieved performance highlights the capability of deep learning models to assist in medical image analysis and improve diagnostic accuracy.

The system not only reduces the dependency on manual ECG interpretation but also enhances the efficiency and reliability of the diagnostic process. Through the use of automated classification, it enables faster analysis of ECG data, which can be particularly beneficial in high-demand healthcare environments. The combination of preprocessing techniques, optimized training strategies, and performance evaluation ensures that the model maintains stability and generalization across unseen data.

Furthermore, the framework demonstrates that AI-driven solutions can serve as supportive tools for healthcare professionals by providing timely and consistent predictions. Its implementation using widely available computational resources makes it accessible and scalable for practical use. Overall, this project highlights the potential of deep learning in advancing medical diagnostics, contributing to early detection, improved patient care, and reduced mortality associated with cardiovascular diseases.

6.2 Future Scope

- Integration with real-time ECG monitoring systems
- Development of mobile or web-based diagnostic applications

- Use of larger and more diverse datasets for improved accuracy
- Implementation of advanced hybrid or ensemble deep learning models
- Clinical validation and deployment in healthcare environments

7. References :

[1] S. Dhandapani, H. Somasundaram, and T. Angamuthu,
"Hybrid deep learning framework for heart disease prediction using ECG signal images," *Scientific Reports*, vol. 15, no. 1, p. 33922, 2025.

[2] M. B. Abubaker and B. Babayigit,
"Detection of cardiovascular diseases in ECG images using machine learning and deep learning methods," *IEEE Transactions on Artificial Intelligence*, vol. 4, no. 2, pp. 373–382, 2022.

[3] N. L. Fitriyani, M. Syafrudin, G. Alfian, and J. Rhee,
"HDPM: An effective heart disease prediction model for a clinical decision support system," *IEEE Access*, vol. 8, pp. 133034–133050, 2020.

[4] S. Mohan, C. Thirumalai, and G. Srivastava,
"Effective heart disease prediction using hybrid machine learning techniques," *IEEE Access*, vol. 7, pp. 81542–81554, 2019.

[5] A. K. Pandey, P. Pandey, K. Jaiswal, and A. K. Sen,
"A heart disease prediction model using decision tree," *IOSR Journal of Computer Engineering (IOSR-JCE)*, vol. 12, no. 6, pp. 83–86, 2013.

[6] M. Kavitha, G. Gnaneswar, R. Dinesh, Y. R. Sai, and R. S. Suraj,
"Heart disease prediction using hybrid machine learning model," in *Proceedings of the 6th International Conference on Inventive Computation Technologies (ICICT)*, IEEE, 2021, pp. 1329–1333.

[7] X. Yuan, J. Chen, K. Zhang, Y. Wu, and T. Yang,
"A stable AI-based binary and multiple class heart disease prediction model for IoMT," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 3, pp. 2032–2040, 2021.

[8] M. M. Ahsan and Z. Siddique,
"Machine learning-based heart disease diagnosis: A systematic literature review," *Artificial Intelligence in Medicine*, vol. 128, p. 102289, 2022.

[9] H. Velandia, A. Pardo, M. I. Vera, and M. Vera,
"Systematic review of artificial intelligence and electrocardiography for cardiovascular disease diagnosis," *Bioengineering*, vol. 12, no. 11, p. 1248, 2025.

[10] E. Partovi, A. Babic, and A. Gharehbaghi,
"A review on deep learning methods for heart sound signal analysis," *Frontiers in Artificial Intelligence*, vol. 7, p. 1434022, 2024.

[11] L. Niu, C. Chen, H. Liu, S. Zhou, and M. Shu,
"A deep-learning approach to ECG classification based on adversarial domain adaptation," in *Healthcare*, vol. 8, p. 437, MDPI, 2020.

[12] X. Liu, H. Wang, Z. Li, and L. Qin,
"Deep learning in ECG diagnosis: A review," *Knowledge-Based Systems*, vol. 227, p. 107187, 2021.

[13] K. Fatema, S. Montaha, M. A. H. Rony, S. Azam, M. Z. Hasan, and M. Jonkman,
"A robust framework combining image processing and deep learning hybrid model to classify cardiovascular diseases using a limited number of paper-based complex ECG images," *Biomedicines*, vol. 10, no. 11, p. 2835, 2022.